Chlorine (Cl) : $[Ne]3s^2, 3p^5$

Potassium has one electron in its valence shell and chlorine has seven electron in its valence shell. By loosing one electron potassium attains the inert gas electronic configuration of argon and becomes a unipositive cation (K^+) and chlorine accepts this electron to become uninegative chloride ion (Cl^-) there by attaining the stable electronic configuration of argon. These two ions combine to form an ionic crystal in which they are held together by electrostatic attractive force. The energy required for the formation of one mole of K^+ is 418.81 kJ (ionization energy) and the energy released during the formation of one mole of Cl^- is -348.56 kJ (electron gain enthalpy). The sum of these two energies is positive (70.25 kJ) However, during the formation of one mole potassium chloride crystal from its constituent ions, 718 kJ energy is released. This favours the formation of KCl and its stability.

Evaluate Yourself

3) Explain the ionic bond formation in MgO and CaF_2

10.4 Coordinate covalent bond

In the formation of a covalent bond, both the combining atoms contribute one electron each and these electrons are mutually shared among them. However, in certain bond formation, one of the combining atoms donates a pair of electrons i.e. two electrons which are necessary for the covalent bond formation, and these electrons are shared by both the combining atoms. These type of bonds are called coordinate covalent bond or coordinate bond. The combining atom which donates the pair of electron is called a donor atom and the other atom an acceptor atom. This bond is denoted by an arrow starting from the donor atom pointing towards the acceptor atom. (Later in coordination compound, we will refer the donor atom as ligand and the acceptor atom as central-metal atom/ion.

For Example, in ferrocyanide ion $[Fe(CN)_6]^{4-}$, each cyanide ion (CN^-) donates a pair of electrons to form a coordinate bond with iron (Fe^{2+}) and these electrons are shared by Fe^{2+} and CN^-.

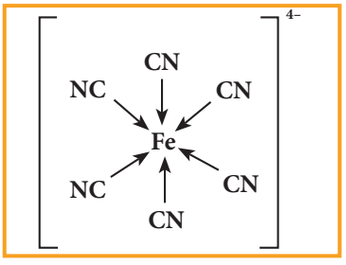


Fig 10. 11 Structure of Ferrocyanide ion

In certain cases, molecules having a lone pair of electrons such as ammonia donates its pair to an electron deficient molecules such as BF_3. to form a coordinate

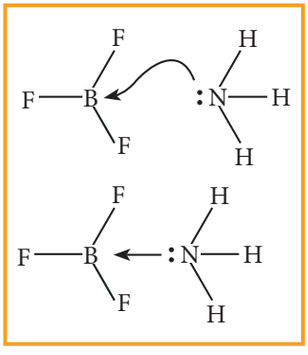


Fig 10. 12 Structure of $BF_3 \rightarrow NH_3$

Unit 10 layout - jagan.indd 75 22-01-2021 12:33:07